ICB response prediction

Nemotron clinical copilot

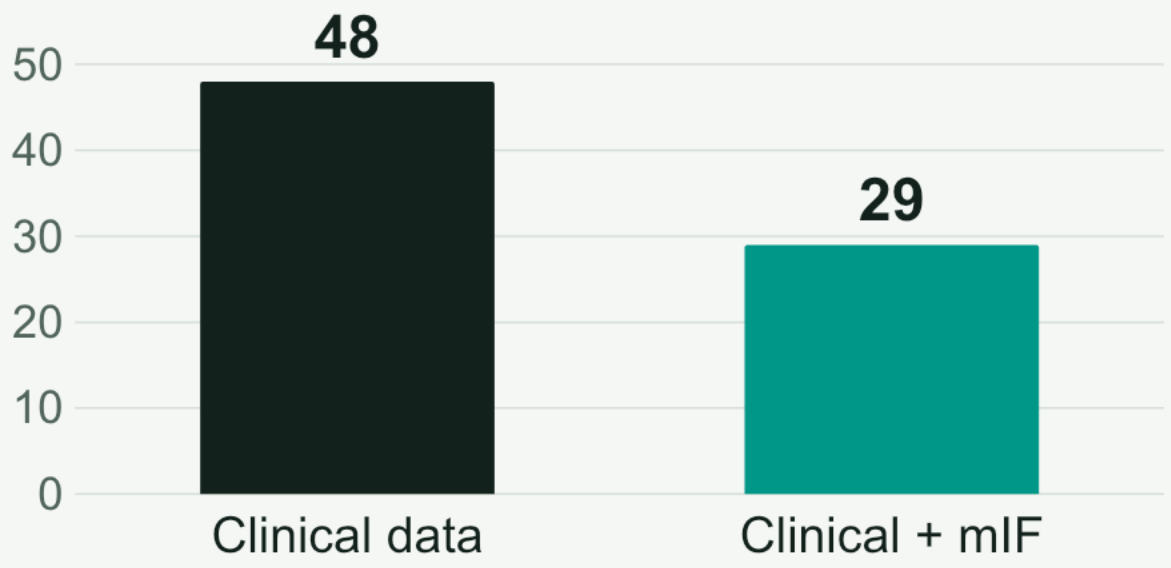
Execution pipeline and model harnesses

Research demonstration



Research scope

Available patients



Research question

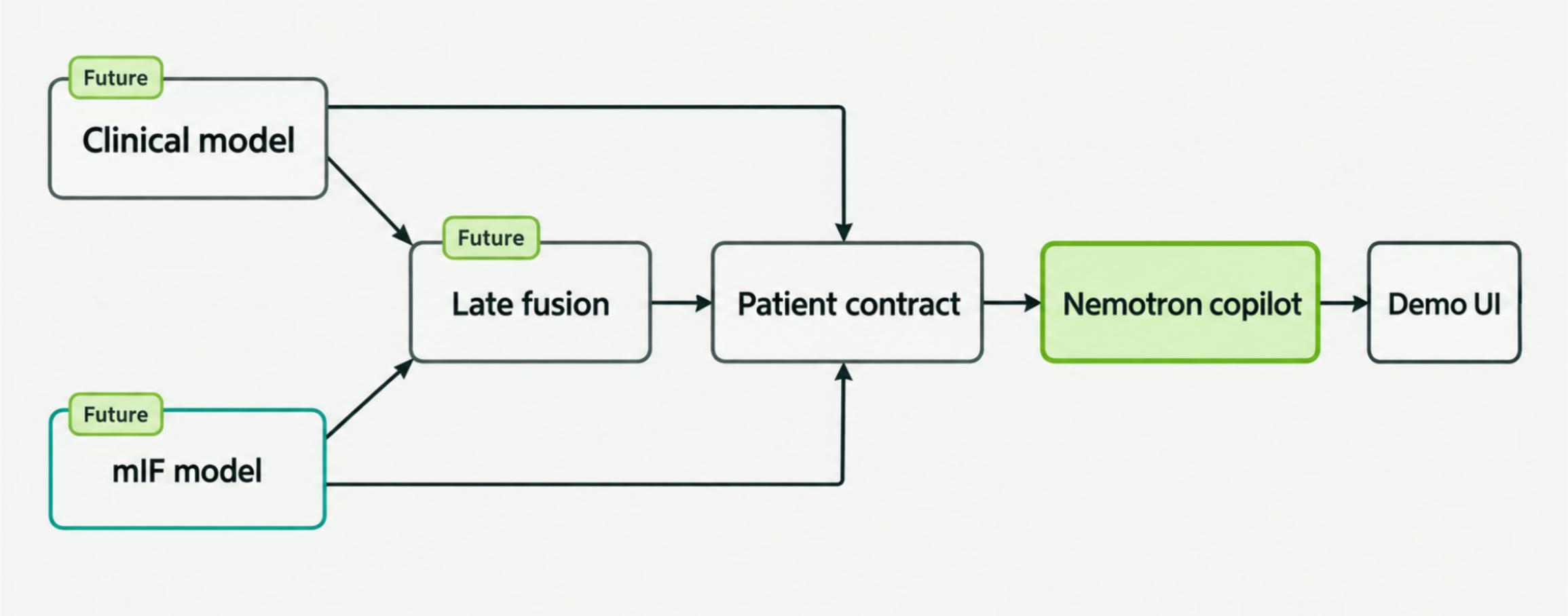
Does combining clinical and mIF information improve prediction of ICB responders?

Planned comparison

Clinical-only, mIF-only and late-fusion predictions on the same evaluation patients.

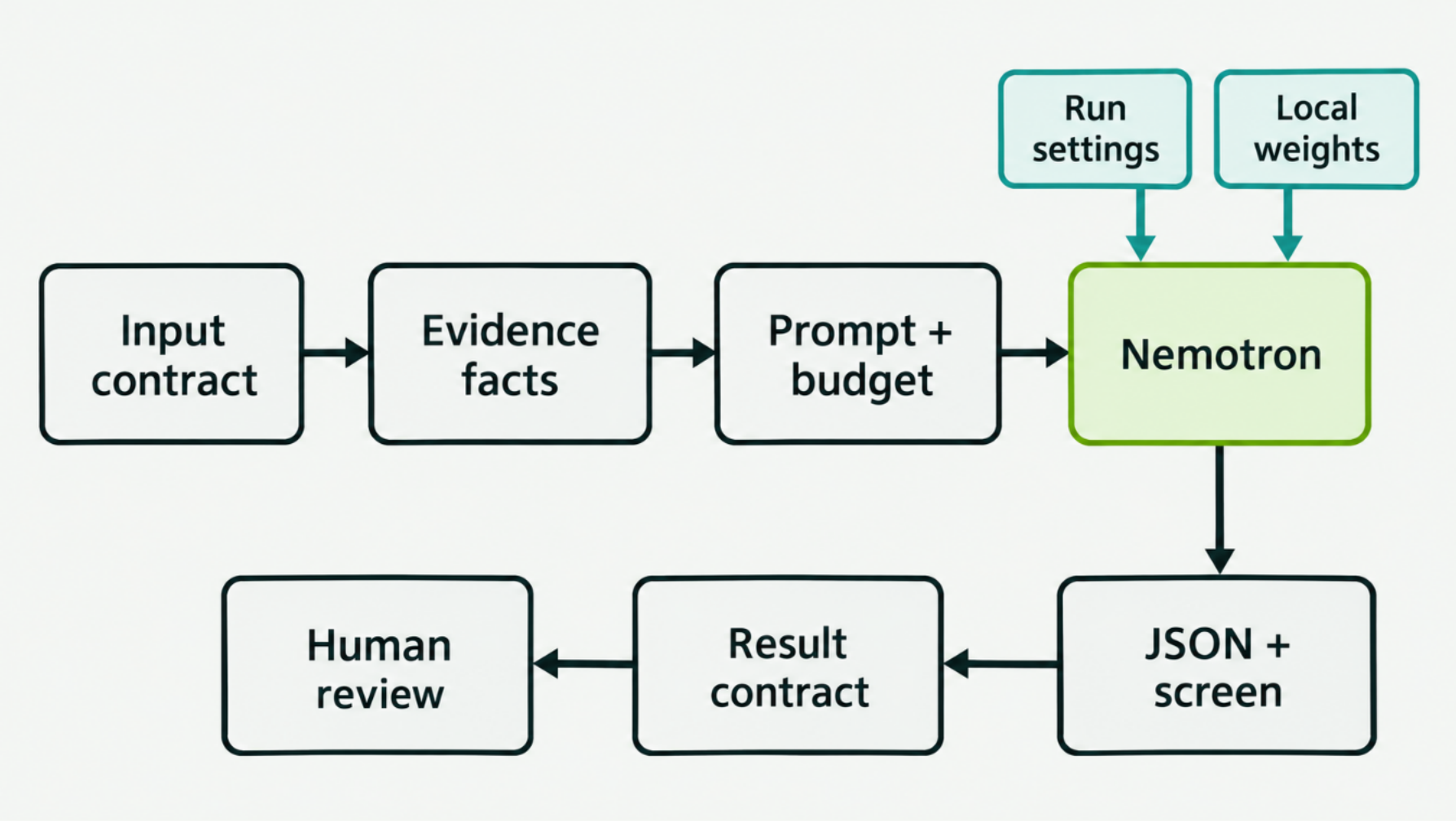
The 29 paired patients are a subset of 48. H&E and extra mIF batches remain possible extensions.

System architecture



Models export scores, features and context. This repository implements the copilot and its interfaces.

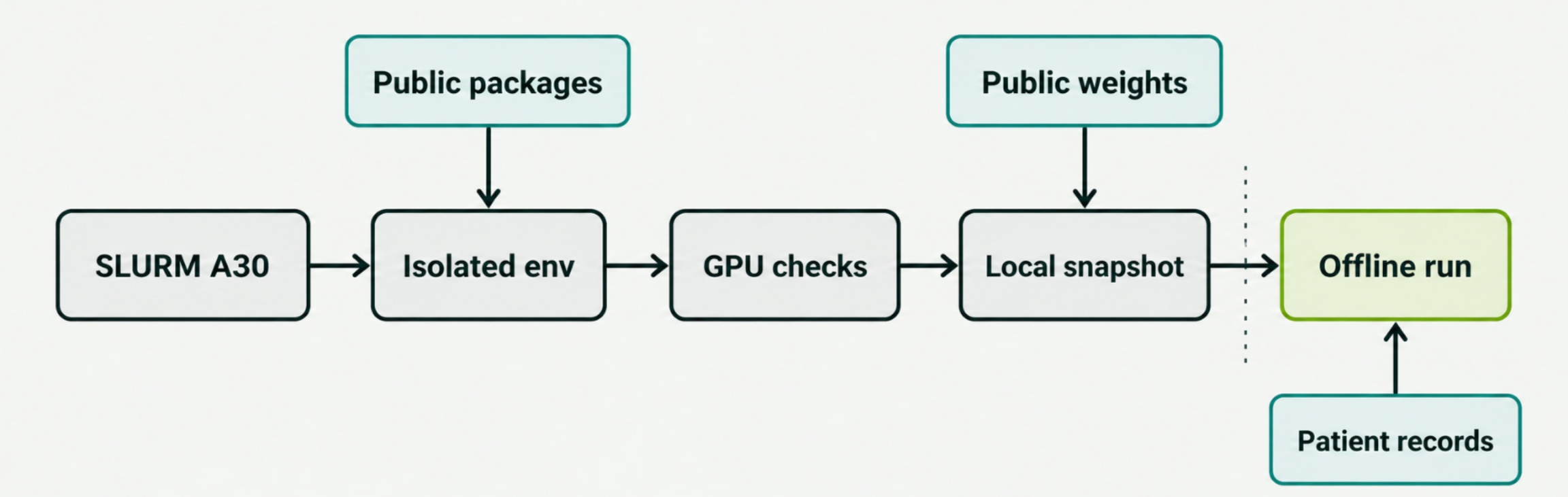
Application harness around Nemotron



The harness controls evidence and execution. Every summary requires independent human review.

HPC execution pipeline

```
bash run_hpc.sh
```



Preparation downloads public artifacts. Patient inference then uses local weights with offline settings.

The GPU engine loads lazily on the first interpretation request.

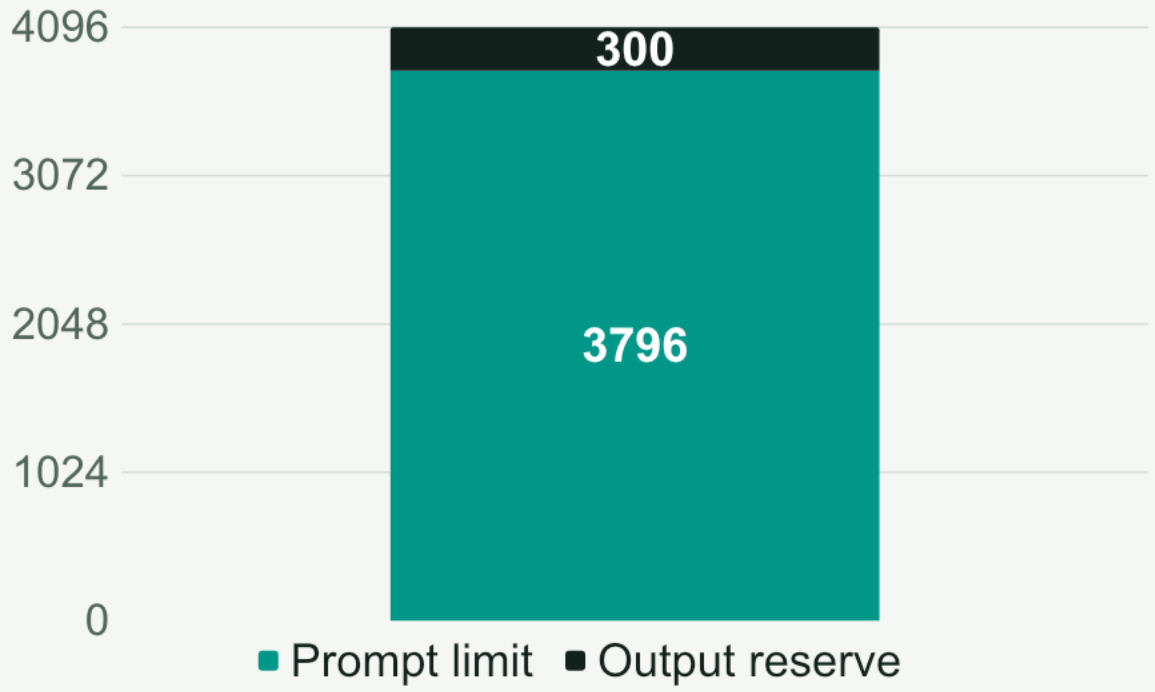
Runtime prerequisites

Layer	Selected deployment requirement
GPU	One A30 with 24 GB VRAM and BF16 support.
Host platform	Linux x86_64 with glibc 2.28 or newer.
NVIDIA driver	570.26 or newer for the selected native CUDA 12.8 path.
Python	Fresh Python 3.11 environment from the Anaconda module.
GPU libraries	vLLM 0.10.2 and PyTorch 2.8.0+cu128.
Tokenizer libraries	Transformers 4.55.2 and Hugging Face Hub 0.34.4.
Model snapshot	Local configuration, tokenizer, weight files and revision metadata.
Job defaults	32 GB host RAM, four CPUs and a four-hour allocation.

Doctor checks CUDA allocation. A generation experiment must verify model loading.

Token budget and execution settings

Default context allocation



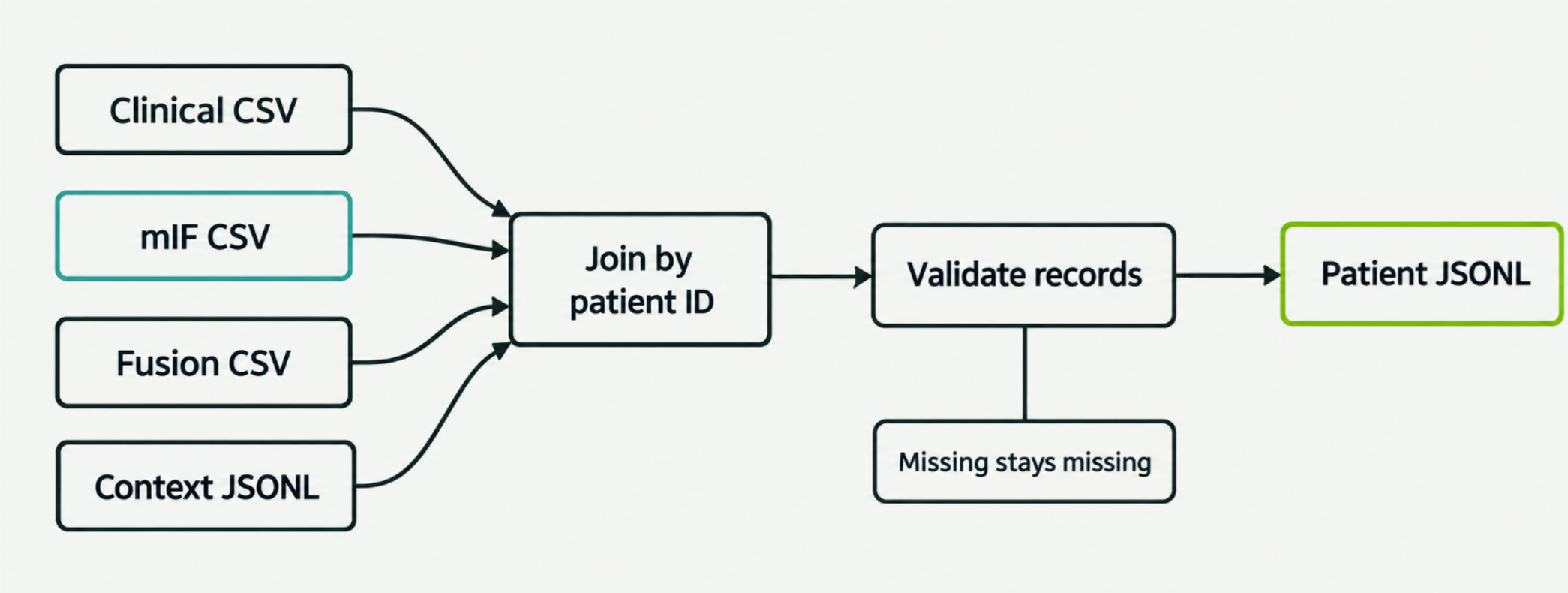
4,096 − 300 = 3,796 input tokens maximum

Current execution defaults

```
dtype = bfloat16
GPU memory fraction = 0.80
GPU concurrent sequences = 4
Experiment batch size = 8
temperature = 0.0
seed = 42
```

Character limits do not guarantee a prompt fits the token budget.

Patient joins and missing modalities



Identity gate

Resolve slide IDs upstream. Export one score per patient per modality.

Availability gate

Missing predictions remain absent. The structured result names unavailable modalities.

Shared patient contract

```
{
  "schema_version": "1.0",
  "patient_id": "P12",
  "predictions": {
    "clinical": {"probability": 0.21},
    "mif": {"probability": 0.78},
    "fused": {"probability": 0.57}
  },
  "response_definition": "Not specified"
}
```

Patient identity

One canonical patient ID and one prediction per patient per modality.

Score metadata

Model version, calibration status and optional threshold travel with each score.

Feature evidence

Measurement direction and signed attribution are separate.

Positive class: ICB responder. The team has not yet agreed the responder definition.

Feature evidence contract

```
{
  "feature": "LDH level",
  "value": 510,
  "unit": "U/L",
  "direction": "elevated",
  "attribution": {
    "method": "shap",
    "value": -0.18,
    "scale": "log_odds"
  }
}
```

Measurement meaning

Value, units and reference direction describe the supplied observation.

Model effect

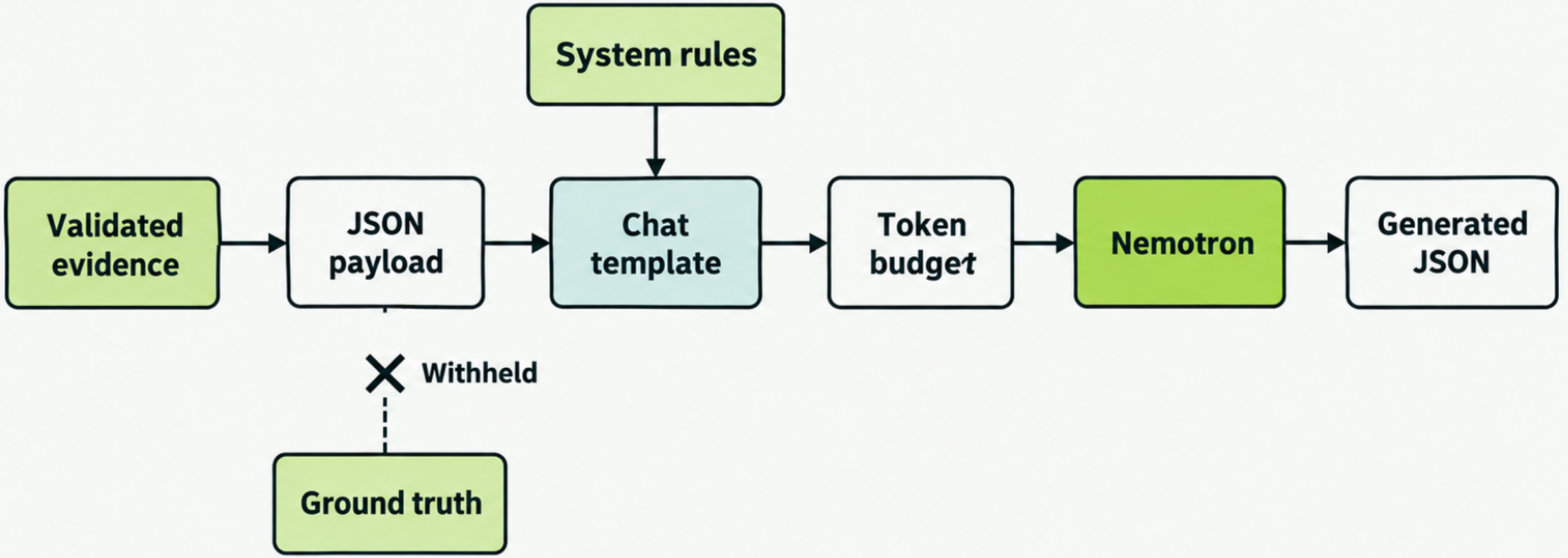
A signed attribution describes that model's responder score on a declared scale.

Upstream semantic checks

Confirm class mapping, units and prediction timepoint. Avoid causal claims.

Synthetic example. A high measurement alone does not determine the feature's effect on a model score.

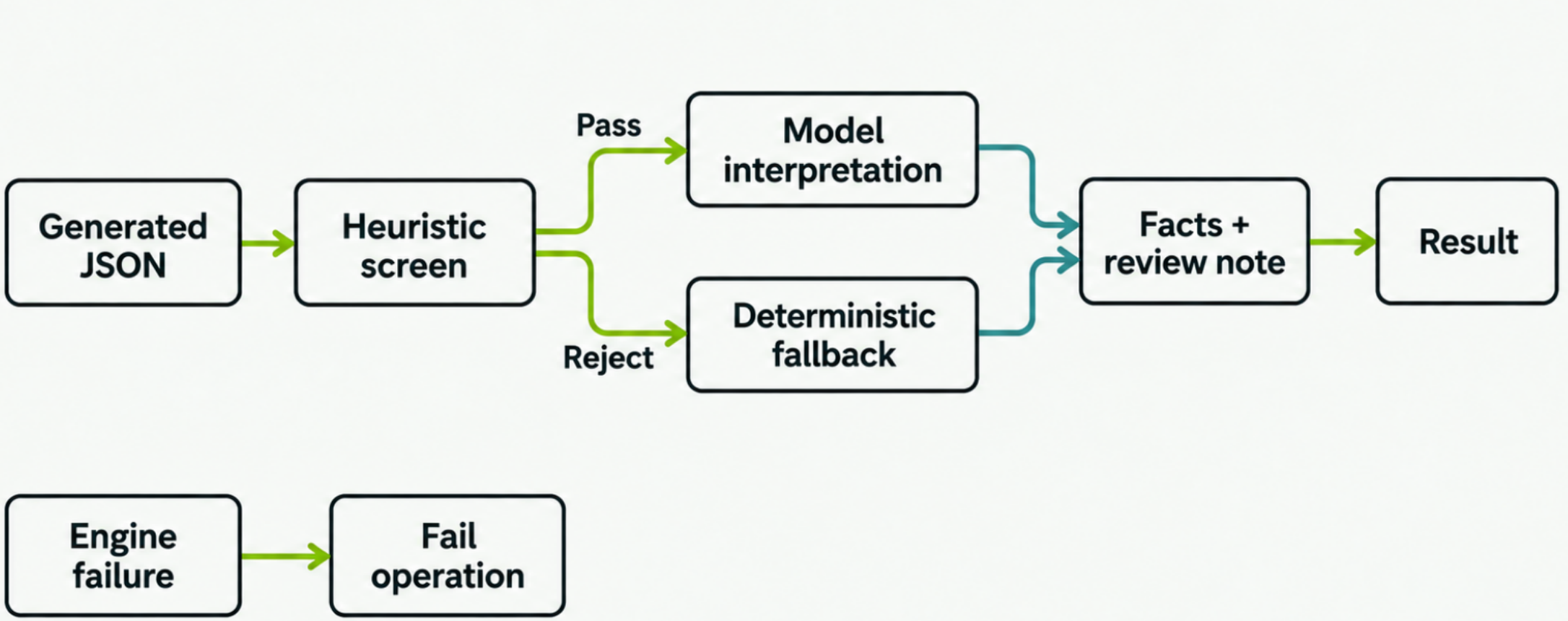
Prompt and context harness




```
Expected model JSON: {"interpretation": "Two concise research sentences."}
```

Ground truth stays outside generation. Human review still checks factual correctness and context leakage.

Output quality gates and fallback



Rejected prose falls back. Engine errors stop a batch run or return an API error.

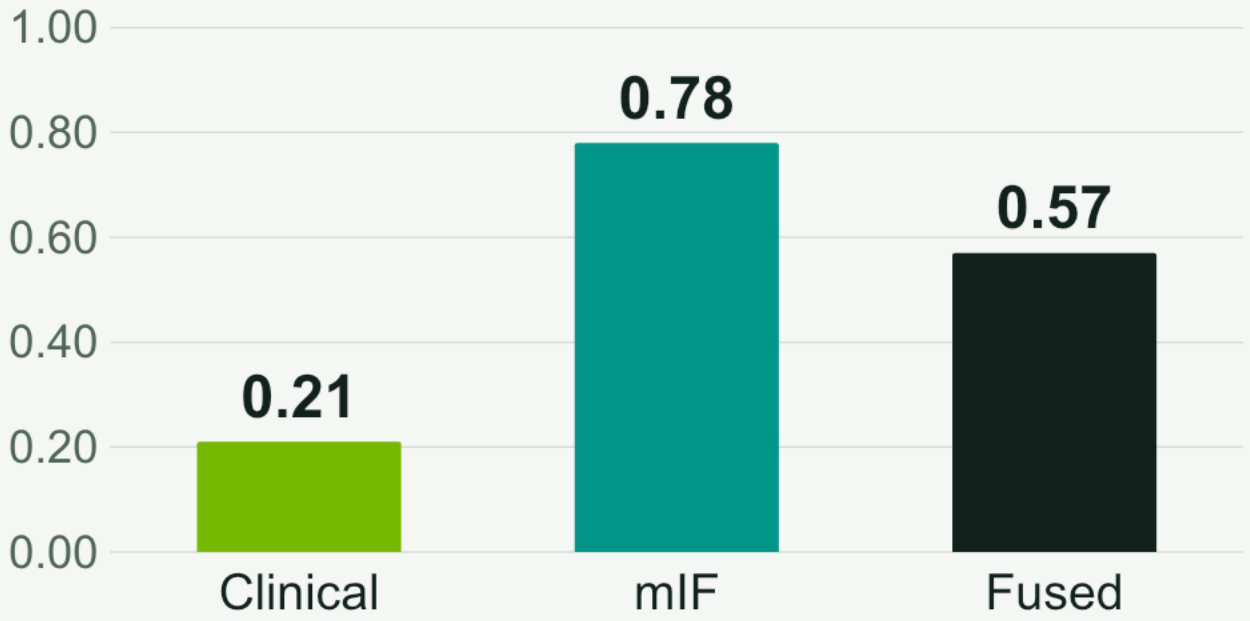
Expected result contract

Field group	Expected behavior
Identity and predictions	Same patient ID and validated scores. No fabricated missing score.
Narrative	Facts from input, a screened interpretation or fallback, and a review note.
Source and quality	Source: mock / vllm / fallback. Quality: mock / passed / fallback.
Comparison and limitations	Show availability, score gaps and supplied-threshold classifications.
Review status	review_required is always true. The summary remains research output.
Provenance	Record prompt version, backend, local model metadata and settings.

Result fields are implemented. A formal response-schema validator is a proposed hardening step.

Synthetic case: discordant predictions

P12 / synthetic model scores



“Clinical and spatial mIF predictions are discordant and warrant review of the supplied evidence.”

Clinical–mIF gap: 0.57
Calibration: unknown
Human review: required

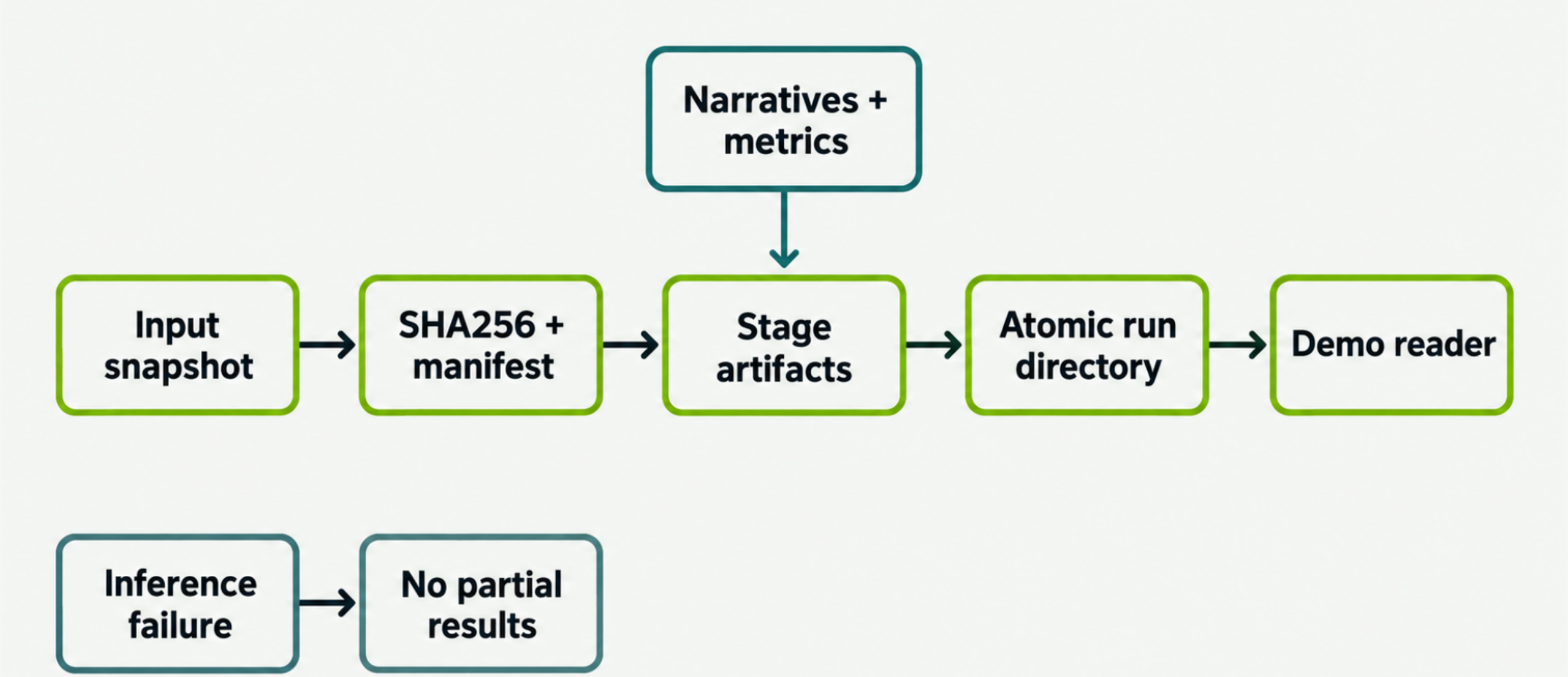
This is deterministic mock output. The supplied fused score does not establish better performance.

Failure handling and recovery

Failure	Current behavior	Operator response
Contract error	Reject input before generation.	Correct IDs, types or metadata.
Prompt too long	Reject before token generation.	Reduce approved context or features.
GPU / cache failure	Stop runtime or scheduled job.	Inspect doctor, cache and SLURM logs.
Rejected model prose	Publish deterministic fallback.	Inspect screen issues and review the case.
Engine failure	Stop run. API returns an error.	Fix the runtime and retry in a new run dir.
Existing result directory	Protect the completed run.	Choose a new output directory.

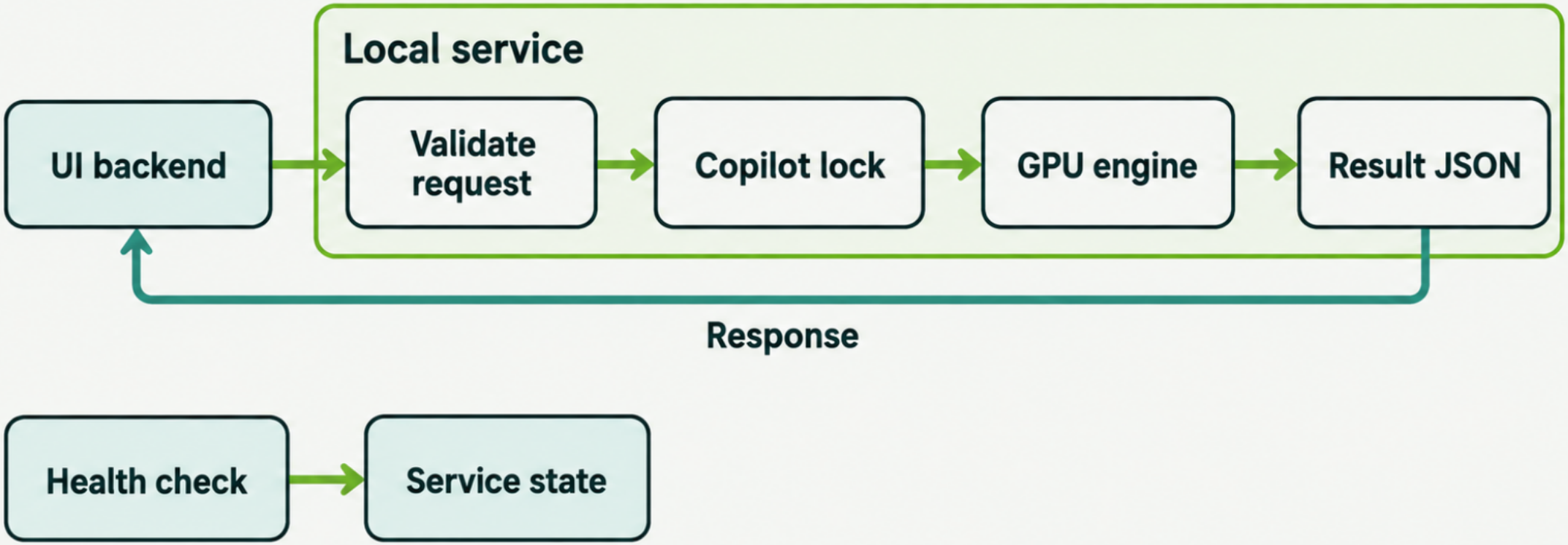
A fallback is a labeled summary. A runtime failure is an execution error, with no partial batch publication.

Experiment publication and provenance



narratives.jsonl narratives.csv report.md manifest.json metrics.json

Local API execution envelope



Current request limits

2 MB requests. Batches contain 1–128 patients. Model calls serialize.

Readiness distinction

Health checks service state. Generation verifies model execution.

Authentication, bounded queues and inference deadlines are proposed additions, not current API features.

Testing harness and readiness gates



Implemented tests

Schema, adapters, mocks, artifacts, HTTP and mocked SLURM tests pass.

Remaining site checks

Run four synthetic cases on A30. Review each resulting narrative.

Software tests verify the harness. Prediction and narrative quality need separate evidence.

Acceptance contracts for the real-data run

Gate	Evidence required before demo use
Identity and completeness	Expected IDs and row counts match. Missing modalities stay explicit.
Prediction meaning	Responder class, endpoint, units and attribution scales are confirmed.
Runtime execution	Real A30 generation succeeds and source/fallback counts are visible.
Narrative review	Review numeric facts, supplied evidence and unsupported claims.
Model evaluation	Compare eligible paired patients using agreed folds and endpoint.
Demo display	Show scores, source, limitations and the human-review flag.

Proposed acceptance rubric. The team must agree fallback tolerance, latency targets and the reviewer.

Connections to future models

Owner / workstream	Handoff to the copilot
Isis / clinical model	Clinical prediction CSV plus clinical features and context.
Alyssa / mIF model	mIF prediction CSV plus feature metadata and canonical patient IDs.
Fusion owner / TBD	Fused prediction CSV with model provenance for paired patients.
Paulo / copilot	Assemble, validate and interpret the combined patient records.
Junhao / demo UI	Read structured narrative results or call the local interpretation API.

Future H&E predictions require a versioned contract and adapter change. Raw images stay upstream.

Evaluation and review

Prediction quality

Use the same eligible patients across modalities.

Agree the endpoint and preserve patient-level folds.

Inspect AUROC and Brier score on paired outputs.

Narrative quality

Check reproduced scores and supplied features.

Review disagreement and missing modalities.

Inspect unsupported claims, source and limitations.

A successful mock run establishes interface behavior. Clinical validity and multimodal benefit require further evidence.

Study and evaluation limitations

Limitation	Implication and required action
Unspecified endpoint	Responder definition remains undecided. Agree the assessment window before evaluation.
Small paired cohort	29 patients have mIF. The eligible comparison may shrink after exclusions.
Uneven mIF availability	The paired group may differ from other patients. Describe the missingness.
Patient or fold leakage	Keep patient slides together. Fit preprocessing and fusion within training folds.
Changing assays or cohort	Extra mIF batches have reported batch effects. H&E needs a new modality contract.

Current metrics are exploratory point estimates. Confidence intervals and significance tests are absent.

Score interpretation edge cases

Synthetic example with a supplied threshold of 0.5

Structured score	Narrative display	Class at threshold 0.5
0.4996	0.500	0 (score below threshold)
0.5004	0.500	1 (score at or above threshold)

Full precision matters

Both scores print as 0.500. The UI should use the structured score and supplied threshold.

Calibration and uncertainty

Calibration is producer metadata. Score gaps and agreement do not quantify confidence.

Missing thresholds remain explicit. The copilot does not infer a default classification threshold.

Patient and evidence edge cases

Edge case	Current behavior and review action
Missing mIF or fusion	The score stays absent. Show availability and avoid inventing a fusion result.
ID and context errors	Duplicate IDs and unmatched context records fail. Resolve slide aggregation upstream.
Reversed responder class	A valid numeric score can target the wrong class. Confirm class order.
Units or attribution signs	Types can pass while meaning is wrong. Verify units, target class and scale.
Outcome text in context	Ground truth is withheld explicitly. Review free text for future-outcome leakage.

Input validation checks field structure. Producers confirm the clinical and study meaning.

Narrative and review limitations

General instruction model	NVIDIA documents risks of inaccurate, biased or incomplete answers. Clinical narrative quality remains unverified.
Lexical output screen	Unsupported paraphrases can pass. Harmless negated phrases can trigger a fallback.
Evidence and fallback	Facts preserve supplied values, including upstream errors. Deterministic fallback also requires human review.

Measure the added value of generated prose against the deterministic template.

Runtime and publication edge cases

Trigger	Current behavior and remaining control
Healthy but cold service	Health can pass before model loading. Add a generation readiness probe.
Context overflow or GPU OOM	The operation fails. Measure token and VRAM headroom on the A30.
Busy GPU or caller retry	Calls serialize. No bounded queue, inference deadline or retry deduplication.
Killed job or storage error	Ordinary errors avoid partial publication. Checkpoints and crash durability are absent.
Sensitive context or API access	Context remains in artifacts. Offline flags do not provide isolation or authentication.

These operational limits require deployment checks. The current runner does not add these controls.

Deployment trade-offs

Current choice	Benefit	Cost or decision
Local Nemotron 4B	Targets one local GPU.	Narrative quality needs review. Larger-model benefit remains untested.
Unquantized BF16	Explicit runtime and precision.	Measure VRAM first. Quantization needs fresh validation.
Bounded evidence	Limits prompt size and excess text.	Owners select evidence. Long prompts fail instead of truncating.
Strict canonical contract	Exposes malformed exports and joins.	New fields or H&E require explicit adapters and versioning.

Compare alternatives after defining review criteria. No performance or quality advantage is assumed.

Demo and modeling trade-offs

Choice	Benefit	Cost or decision
Screened prose or templates	Prose may improve readability.	Compare added value. Both paths still require evidence review.
Completed files or live API	Files provide a stable reviewed demo.	Live interaction needs cold-load and overload testing.
Paired or available cohorts	Paired scores use identical patients.	Smaller selected cohort. Report clinical-only results separately.
Simple or complex fusion	A simple baseline limits tuning.	Assign an owner. Complexity needs valid folds and more evidence.

Proposed hackathon approach: reviewed files, an explicit fusion baseline and exploratory paired evaluation.

Edge-case acceptance checks

Proposed checks before accepting the real-data demo

Scenario	Expected acceptance evidence
Wrong class, unit or timepoint	Producer review catches semantic errors before release.
Missing score or rounded tie	The UI shows availability, adequate precision and supplied thresholds.
No eligible labels or one class	Metrics keep exclusions explicit. Undefined AUROC stays null.
Passed or fallback narrative	Review unsupported claims and omissions. Record source and rejection issues.
Cold, long or busy GPU request	Real A30 checks expose readiness, failures and agreed latency limits.

Proposed acceptance cases. Real GPU execution and independent narrative review remain pending.

Harness hardening still to add

Proposed additions beyond the current implementation

Harness area	Addition to design and implement
Machine contracts	Versioned JSON Schemas and output cross-field invariants.
Evidence semantics	Check endpoint, timepoint, units, class mapping and fold provenance.
Live service controls	Model warmup, bounded queue, inference deadline and access policy.
Local observability	Request IDs, stage latency, live fallback counters and failed-run records.
Reproducibility	Hash-locked environment and weight digests when needed.
Acceptance automation	GPU regression cases plus independent narrative review.

These tasks strengthen execution and auditability. They do not establish clinical validity by themselves.

Decisions needed for the real-data demo

Proposed starting choices and owners for team confirmation

Decision	Suggested owner	Starting choice to confirm
Responder endpoint	Clinical owner TBD + Isis	Agree the responder definition and assessment window.
Patient identity	Isis + Alyssa	Use canonical IDs. Resolve slide aggregation upstream.
Evaluation design	Modeling team	Share patient-level folds. Start with the paired cohort.
Late-fusion method	Owner TBD	Assign an owner. Prespecify a simple fusion baseline.
Export metadata	All model owners	Fix class mapping, attribution scales and score metadata.
Demo and acceptance	Paulo + Junhao Reviewer TBD	Choose demo mode. Verify A30 output and review cases.

H&E and extra mIF batches also need a scope decision before inclusion.

Completion checkpoints

Checkpoint	Evidence of completion
Contract agreement	The team records the endpoint, IDs, splits and model exports.
A30 smoke experiment	Paulo generates local Nemotron output from the synthetic cases.
Real prediction handoff	Model owners supply held-out clinical, mIF and fused scores.
Review and comparison	The team inspects paired metrics and representative narratives.
Demo integration	Junhao displays each patient’s scores, evidence and review status.

Ready now: mock experiment, HPC runner and integration interfaces. Pending: real GPU and model handoffs.